

Practice Questions for Auxiliary Problem

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December 31, 2024

1 Questions

1. Why is the auxiliary problem necessary when the origin is not feasible in a linear programming problem?
 - a. The auxiliary problem provides a way to graphically solve the linear program.
 - b. The auxiliary problem transforms the problem into a dual problem, which is easier to solve.
 - c. The auxiliary problem ensures that the simplex method can start from a feasible solution.
 - d. The auxiliary problem simplifies the objective function, making it easier to optimize.
 - e. The auxiliary problem eliminates redundant constraints from the original problem.
2. In the auxiliary problem, what is the significance of the optimal objective function value being zero?
 - a. It indicates that the auxiliary problem is unbounded.
 - b. It means the original problem is infeasible.
 - c. It implies that the original problem has a feasible solution.
 - d. It signifies that the auxiliary problem has multiple optimal solutions.
 - e. It suggests that the original problem's objective function is linear.
3. How does the auxiliary problem help in finding an initial feasible solution for the simplex method when the origin is infeasible?
 - a. It directly provides the optimal solution to the original problem.
 - b. It introduces a new variable that shifts the feasible region to include the origin.
 - c. It creates a modified problem with a guaranteed feasible solution, allowing the simplex method to find a feasible solution for the original problem.
 - d. It simplifies the constraints of the original problem, making it easier to find a feasible solution.
 - e. It changes the objective function to a linear function, making it easier to solve.
4. What is the interpretation of a negative value for a slack variable in the auxiliary problem before finding a feasible solution?
 - a. It indicates that the corresponding constraint in the original problem is satisfied.
 - b. It means the corresponding constraint in the auxiliary problem is redundant.
 - c. It shows that the corresponding constraint in the original problem is violated.

- d. It implies that the original problem is unbounded.
 - e. It signifies that the auxiliary problem has reached its optimal solution.
5. Consider a linear programming problem where the origin (0,0) is not a feasible solution. Explain the purpose of the auxiliary problem in finding a feasible solution for the simplex method, and describe how the solution to the auxiliary problem helps in solving the original problem.
- a. The auxiliary problem introduces artificial variables to convert inequality constraints into equalities, allowing the simplex method to proceed directly from the origin.
 - b. The auxiliary problem identifies redundant constraints in the original problem, simplifying it before applying the simplex method.
 - c. The auxiliary problem finds a feasible solution to the original problem by introducing a new objective function and constraints that guarantee feasibility, providing a starting point for the simplex method on the original problem.
 - d. The auxiliary problem computes the gradient of the objective function to determine the direction of steepest ascent, guiding the simplex method towards a feasible solution.
 - e. The auxiliary problem is used to check the feasibility of the original problem; if the origin is infeasible, the auxiliary problem is not needed.
6. In the auxiliary problem, what is the significance of the optimal objective function value being zero?
- a. It indicates that the original problem is unbounded.
 - b. It means the auxiliary problem is infeasible.
 - c. It signifies that the original problem is feasible.
 - d. It implies that the original problem has multiple optimal solutions.
 - e. It suggests that the original problem's objective function is not well-defined.
7. In the auxiliary problem, what is the interpretation of a negative value for a slack variable before finding a feasible solution?
- a. It indicates an unbounded solution.
 - b. It signifies that the corresponding constraint is violated.
 - c. It means the auxiliary problem is optimal.
 - d. It implies that the original problem is infeasible.
 - e. It suggests that a non-basic variable should leave the basis.
8. Consider the auxiliary problem: $\max \xi = -x_0$ subject to $w_1 = -1 + x_1 - x_2 + x_0$, $w_2 = -2 + x_1 + 2x_2 + x_0$, $w_3 = 1 - x_2 + x_0$, $x_1, x_2, x_0, w_1, w_2, w_3 \geq 0$. If we set $x_1 = x_2 = x_0 = 0$, which constraint is most violated?
- a. $w_1 \geq 0$
 - b. $w_2 \geq 0$
 - c. $w_3 \geq 0$

- d. All constraints are satisfied.
 - e. The objective function is maximized.
9. After solving the auxiliary problem and obtaining $x_0 = 0$, how is this solution used to find a feasible solution for the original problem?
- a. The values of x_0 , x_1 , and x_2 directly form a feasible solution for the original problem.
 - b. The values of x_1 and x_2 are substituted into the original problem's objective function.
 - c. The value of x_0 is substituted into the equations obtained from the auxiliary problem to eliminate x_0 , yielding a feasible solution for the original problem's variables.
 - d. The auxiliary problem's objective function value is added to the original problem's objective function value.
 - e. The solution is infeasible; the auxiliary problem failed to find a feasible solution for the original problem.
10. Consider the auxiliary problem presented on Slide 33: $\max \xi = -x_0$ subject to $w_1 = -1 + x_1 - x_2 + x_0$, $w_2 = -2 + x_1 + 2x_2 + x_0$, $w_3 = 1 - x_2 + x_0$, $x_1, x_2, x_0, w_1, w_2, w_3 \geq 0$. If we set $x_1 = x_2 = 0$, what is the value of x_0 that makes $w_2 = 0$?
- a. 0
 - b. 1
 - c. 2
 - d. 3
 - e. 4

2 Answers

1. Why is the auxiliary problem necessary when the origin is not feasible in a linear programming problem?
 - a. The auxiliary problem is an algebraic technique, not a graphical one. Graphical methods are used for visualizing the feasible region but not directly for solving the auxiliary problem.
 - b. The auxiliary problem does not transform the problem into its dual. The dual problem is a different linear program related to the original one, but the auxiliary problem is a modification of the primal problem itself.
 - c. The auxiliary problem introduces a new variable and modifies the constraints to create a new problem that is always feasible, even if the original problem is not. This allows the simplex method, which requires a feasible starting point, to be applied.
 - d. The auxiliary problem's objective function is different from the original problem's objective function. It aims to find a feasible solution, not to directly simplify the original objective function.
 - e. The auxiliary problem does not eliminate constraints. It modifies the constraints to guarantee feasibility.
2. In the auxiliary problem, what is the significance of the optimal objective function value being zero?
 - a. An optimal objective function value of zero does not indicate unboundedness. Unboundedness means the objective function can be made arbitrarily large.
 - b. An optimal objective function value of zero indicates the opposite: that the original problem is feasible. If the original problem were infeasible, the optimal value would be negative.
 - c. If the optimal value of the auxiliary problem is zero, it means that a feasible solution to the original problem exists, and the auxiliary variable x_0 can be set to zero, satisfying all constraints of the original problem.
 - d. While the auxiliary problem might have multiple optimal solutions, a zero optimal value specifically relates to the feasibility of the original problem.
 - e. The linearity of the original objective function is a given condition; the auxiliary problem's significance relates to feasibility, not the linearity of the objective function.
3. How does the auxiliary problem help in finding an initial feasible solution for the simplex method when the origin is infeasible?
 - a. The auxiliary problem does not directly provide the optimal solution to the original problem. It provides a feasible starting point from which the simplex method can be applied to the original problem.
 - b. The auxiliary problem doesn't shift the feasible region. It creates a new problem with a different feasible region that is guaranteed to contain a feasible solution.
 - c. The auxiliary problem modifies the constraints to guarantee a feasible solution, even if the original problem is infeasible. The simplex method is then used to solve this modified problem, and the solution is used to find a feasible solution for the original problem.
 - d. The auxiliary problem doesn't simplify the constraints; it modifies them to ensure feasibility.
 - e. The objective function of the auxiliary problem is different from the original problem's objective function. The original problem's objective function's linearity is assumed.

4. What is the interpretation of a negative value for a slack variable in the auxiliary problem before finding a feasible solution?
 - a. A negative slack variable means the constraint is **not** satisfied.
 - b. A negative slack variable doesn't indicate redundancy. Redundancy means a constraint is unnecessary because it's implied by others.
 - c. A negative slack variable indicates that the corresponding constraint in the original problem is violated. The auxiliary problem aims to resolve these violations by adjusting the auxiliary variable.
 - d. A negative slack variable doesn't imply unboundedness. Unboundedness refers to an objective function that can increase infinitely.
 - e. A negative slack variable is a sign that the auxiliary problem is still in progress and hasn't reached its optimal solution.
5. Consider a linear programming problem where the origin (0,0) is not a feasible solution. Explain the purpose of the auxiliary problem in finding a feasible solution for the simplex method, and describe how the solution to the auxiliary problem helps in solving the original problem.
 - a. The auxiliary problem does not introduce artificial variables to convert inequalities to equalities. Artificial variables are used in a different technique (two-phase method) to handle infeasibility, but the auxiliary problem's approach is different.
 - b. The auxiliary problem does not focus on identifying redundant constraints. Redundant constraints are dealt with separately in pre-processing steps, not as part of the auxiliary problem's core function.
 - c. The auxiliary problem is designed to find a feasible solution for the original problem when the origin is infeasible. It does this by introducing a new objective function and constraints that guarantee feasibility. The solution to this auxiliary problem then provides a starting point for the simplex method to be applied to the original problem.
 - d. The simplex method does not use gradient calculations. It's an iterative algorithm that moves from one feasible solution to another, improving the objective function at each step.
 - e. The auxiliary problem is precisely needed when the origin is infeasible. It provides a way to find a feasible starting point for the simplex method.
6. In the auxiliary problem, what is the significance of the optimal objective function value being zero?
 - a. An unbounded original problem would not necessarily lead to an optimal objective function value of zero in the auxiliary problem. The auxiliary problem is designed to find a feasible solution, not to determine boundedness.
 - b. If the auxiliary problem is infeasible, it means the original problem is also infeasible. An optimal objective function value of zero indicates feasibility.
 - c. An optimal objective function value of zero in the auxiliary problem means that a feasible solution to the original problem exists, because x_0 can be set to zero, satisfying all constraints of the original problem.
 - d. The auxiliary problem's objective function value doesn't directly relate to the number of optimal solutions in the original problem. It only determines feasibility.
 - e. The objective function of the original problem is assumed to be well-defined; the auxiliary problem is a tool to find a feasible starting point for solving it.

7. In the auxiliary problem, what is the interpretation of a negative value for a slack variable before finding a feasible solution?
 - a. Unboundedness is a different concept related to the objective function's behavior, not the constraint violation.
 - b. A negative slack variable value means the corresponding constraint in the auxiliary problem is not satisfied. This is an indication of infeasibility in the current iteration.
 - c. A negative slack variable indicates infeasibility, not optimality. Optimality is reached when all slack variables are non-negative and the objective function is maximized.
 - d. While a negative slack variable shows a problem, it doesn't definitively prove the original problem is infeasible. The auxiliary problem aims to find a feasible solution.
 - e. The decision of which variable leaves the basis depends on the simplex method's rules, not solely on a negative slack variable.

8. Consider the auxiliary problem: $\max \xi = -x_0$ subject to $w_1 = -1 + x_1 - x_2 + x_0$, $w_2 = -2 + x_1 + 2x_2 + x_0$, $w_3 = 1 - x_2 + x_0$, $x_1, x_2, x_0, w_1, w_2, w_3 \geq 0$. If we set $x_1 = x_2 = x_0 = 0$, which constraint is most violated?
 - a. While w_1 is negative, w_2 is more negative, indicating a greater violation.
 - b. With $x_1 = x_2 = x_0 = 0$, we have $w_1 = -1$, $w_2 = -2$, and $w_3 = 1$. $w_2 = -2$ represents the most significant violation of the non-negativity constraint.
 - c. w_3 is positive and satisfies the constraint.
 - d. The constraints $w_1 \geq 0$ and $w_2 \geq 0$ are violated.
 - e. The objective function is not maximized; a negative value of x_0 would indicate a better solution.

9. After solving the auxiliary problem and obtaining $x_0 = 0$, how is this solution used to find a feasible solution for the original problem?
 - a. The auxiliary problem introduces x_0 ; only x_1 and x_2 are relevant to the original problem.
 - b. While the values of x_1 and x_2 are eventually used in the original objective function, the key step is first obtaining a feasible solution.
 - c. Substituting $x_0 = 0$ into the equations for x_1 and x_2 (obtained during the solution of the auxiliary problem) eliminates x_0 and provides values for x_1 and x_2 that satisfy the original problem's constraints.
 - d. The objective functions are separate; the auxiliary problem's solution only helps find a feasible starting point for the original problem.
 - e. An x_0 value of 0 indicates success; the auxiliary problem found a feasible solution for the original problem.

10. Consider the auxiliary problem presented on Slide 33: $\max \xi = -x_0$ subject to $w_1 = -1 + x_1 - x_2 + x_0$, $w_2 = -2 + x_1 + 2x_2 + x_0$, $w_3 = 1 - x_2 + x_0$, $x_1, x_2, x_0, w_1, w_2, w_3 \geq 0$. If we set $x_1 = x_2 = 0$, what is the value of x_0 that makes $w_2 = 0$?
 - a. Answer A is incorrect because if $x_0 = 0$, then $w_2 = -2$, which is not feasible.
 - b. Answer B is incorrect because if $x_0 = 1$, then $w_2 = -1$, which is not feasible.

- c. Answer C is correct. Setting $x_1 = x_2 = 0$ in the constraint $w_2 = -2 + x_1 + 2x_2 + x_0$ gives $w_2 = -2 + x_0$. To make $w_2 = 0$, we need $x_0 = 2$.
- d. Answer D is incorrect because if $x_0 = 3$, then $w_2 = 1$, which is feasible, but we are looking for the value of x_0 that makes $w_2 = 0$.
- e. Answer E is incorrect because if $x_0 = 4$, then $w_2 = 2$, which is feasible, but we are looking for the value of x_0 that makes $w_2 = 0$.